

Lab 3: Noise in Quantum Computers

QML

1 introduction

real quantum computers r noisy. like, really noisy. every gate u apply has some probability of error, qubits lose their quantum state over time (decoherence), and measurements r not perfectly accurate. this matters a lot for quantum ML bcs our variational circuits will need to run on real hardware eventually.

in this lab u'll learn to:

- simulate different types of noise using qiskit Aer
- compare ideal vs noisy circuit execution
- understand the main error channels (bit flip, phase flip, depolarizing, amplitude damping)
- implement basic error mitigation techniques (ZNE and measurement error mitigation)

Key Point. we r not doing full quantum error correction here (that requires way more qubits). instead, we're doing error mitigation — clever classical post-processing tricks that reduce the impact of noise without adding extra qubits.

2 noise channels

2.1 types of noise

the main noise channels u need to know:

Definition 2.1. the bit flip channel flips $|0\rangle \leftrightarrow |1\rangle$ with probability p . it's described by:

$$\mathcal{E}(\rho) = (1-p)\rho + pX\rho X$$

Definition 2.2. the phase flip channel (dephasing) applies a Z error with probability p :

$$\mathcal{E}(\rho) = (1-p)\rho + pZ\rho Z$$

Definition 2.3. the depolarizing channel replaces the state with the maximally mixed state with probability p :

$$\mathcal{E}(\rho) = (1-p)\rho + \frac{p}{3}(X\rho X + Y\rho Y + Z\rho Z)$$

this is the “everything goes wrong equally” model and it's the most commonly used for benchmarking.

Definition 2.4. the amplitude damping channel models energy loss (like a qubit decaying from $|1\rangle$ to $|0\rangle$). it's described by Kraus operators:

$$K_0 = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-\gamma} \end{pmatrix}, \quad K_1 = \begin{pmatrix} 0 & \sqrt{\gamma} \\ 0 & 0 \end{pmatrix}$$

where γ is the damping probability.

Intuition. bit flip = classical error (like a noisy communication channel). phase flip = purely quantum error (no classical analogue). depolarizing = uniform random error. amplitude damping = energy dissipation (most physically realistic). in practice, real devices have a mix of all of these.

Exercise 1. visualizing noise channels.

1. prepare the state $|+\rangle = H|0\rangle$ and apply each noise channel (bit flip, phase flip, depolarizing) with $p = 0.1$.
2. for each channel, compute the output density matrix using the qiskit Aer density matrix simulator.
3. use qiskit's `plot_bloch_multivector` to visualize the state before and after noise.
4. which noise channel has the biggest effect on the $|+\rangle$ state? why does this make sense?
to add noise in qiskit:
 - `import NoiseModel from qiskit_aer.noise`
 - import the error types: `depolarizing_error`, `paulli_error`, etc.
 - create a noise model and add errors to specific gates
 - pass the noise model to the Aer simulator when u run

3 building noise models

3.1 custom noise models in qiskit

here's the workflow for adding noise:

1. create a `NoiseModel()` object
2. create error objects (e.g., `depolarizing_error(p, num_qubits)`)
3. add them to the noise model: `noise_model.add_all_qubit_quantum_error(error, ['gate_name'])`
4. run ur circuit with the noisy simulator

u can add different error rates to different gates. typically 2-qubit gates (like CNOT) have much higher error rates than single-qubit gates. on real IBM devices, single-qubit error rates r around 10^{-3} to 10^{-4} and CNOT error rates r around 10^{-2} .

Exercise 1. noisy bell state.

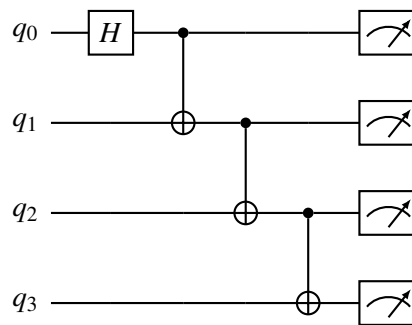
1. build the bell state circuit from lab 1 (H + CNOT + measure)
2. run it with no noise (ideal). record the counts.
3. create a noise model with:
 - depolarizing error $p = 0.01$ on single-qubit gates
 - depolarizing error $p = 0.05$ on CNOT gates

4. run the same circuit with this noise model. record the counts.
5. compare: in the ideal case u should get only 00 and 11. wt extra outcomes appear with noise? wt r their probabilities?
6. increase the CNOT error rate to $p = 0.1, 0.2, 0.3$. plot the probability of getting the “wrong” outcomes (01 or 10) as a function of the error rate.

Exercise 2. noisy GHZ state.

1. build GHZ circuits for $n = 3, 4, 5, 6, 7$ qubits
2. run each one with the realistic noise model (single-qubit error = 0.001, CNOT error = 0.01)
3. for each n , compute the fidelity of the noisy output with the ideal GHZ state
4. plot fidelity vs n . u should see it drop as n increases.
5. explain why more qubits = more noise. roughly how does fidelity scale with the number of CNOT gates?

the circuit for 4-qubit GHZ:



4 amplitude damping and T_1 decay

amplitude damping is the most physically motivated noise model. it represents the qubit losing energy over time (T_1 relaxation).

the probability of decay depends on the gate duration t and the T_1 time:

$$\gamma = 1 - e^{-t/T_1}$$

typical IBM quantum devices have $T_1 \approx 100\mu\text{s}$ and single-qubit gate times of $\approx 35\text{ns}$, so the error per gate is small. but circuits with many gates accumulate error.

Exercise 1. T_1 decay simulation.

1. prepare $|1\rangle$ (apply X to $|0\rangle$)
2. add k identity gates (each representing some time duration)

3. add amplitude damping noise with $\gamma = 0.01$ per gate
4. measure and compute $P(|1\rangle)$
5. repeat for $k = 0, 10, 20, 50, 100, 200$
6. plot $P(|1\rangle)$ vs k . u should see exponential decay.
7. fit an exponential $P = e^{-k/k_0}$ and extract k_0 . compare with $1/\gamma$.

this is essentially how T_1 is measured on real devices!

5 zero-noise extrapolation (ZNE)

ZNE is one of the simplest and most effective error mitigation techniques. the idea is:

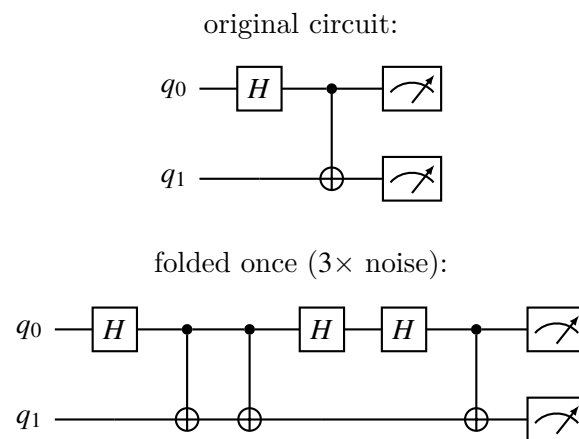
1. run ur circuit at several different noise levels
2. extrapolate back to zero noise

but wait — u can't just turn down the noise on a real device. the trick is to increase the noise by stretching the circuit (adding more gates that don't change the ideal computation but do add noise).

5.1 noise amplification by unitary folding

the simplest way to amplify noise: replace each gate G with $G \cdot G^\dagger \cdot G$. this does the same thing as G (bcs $G^\dagger G = I$) but uses 3x the gates, so it has 3x the noise. u can generalize: $G \rightarrow G(G^\dagger G)^k$ gives a noise scale factor of $2k + 1$.

here's wt unitary folding looks like for a simple circuit:



Exercise 1. implementing ZNE.

1. build a simple circuit: prepare $|+\rangle$ and measure the expectation value of Z (should be 0 ideally).
2. add depolarizing noise with $p = 0.05$ on all gates.

3. run the circuit normally (noise factor = 1). record $\langle Z \rangle$.
4. fold the circuit once (noise factor = 3): for each gate G , replace it with $G \cdot G^\dagger \cdot G$. record $\langle Z \rangle$.
5. fold twice (noise factor = 5). record $\langle Z \rangle$.
6. now extrapolate to noise factor = 0:
 - u have data points at noise factors 1, 3, 5
 - fit a line (or quadratic) through these points
 - evaluate the fit at noise factor = 0
7. compare ur extrapolated value with the ideal value. how much closer r u?

hint for folding in qiskit: create a new circuit. for each gate in the original circuit, add the gate, then its inverse, then the gate again. the inverse of H is H . the inverse of $CNOT$ is $CNOT$. the inverse of $R_y(\theta)$ is $R_y(-\theta)$.

Exercise 2. ZNE on a bell state. apply ZNE to the bell state circuit:

1. measure $\langle ZZ \rangle$ on the bell state (should be +1 ideally)
2. run at noise factors 1, 3, 5, 7
3. extrapolate to zero noise using:
 - linear fit (using noise factors 1 and 3 only)
 - quadratic fit (using noise factors 1, 3, and 5)
4. which fit gives a better estimate? plot ur data points and both fits.
5. wt happens if u try noise factors that r too large (e.g., 15, 21)? why does ZNE break down at very high noise?

6 measurement error mitigation

another common source of error: the measurement itself isn't perfect. on real hardware, u might measure $|0\rangle$ when the qubit is actually in $|1\rangle$, or vice versa.

the idea behind measurement error mitigation:

1. characterize the measurement errors by preparing known states and measuring them
2. build a calibration matrix M where $M_{ij} = P(\text{measure } i \mid \text{prepared } j)$
3. invert this matrix to correct future measurements: $\vec{p}_{\text{corrected}} = M^{-1} \vec{p}_{\text{measured}}$

for 1 qubit, the calibration matrix is:

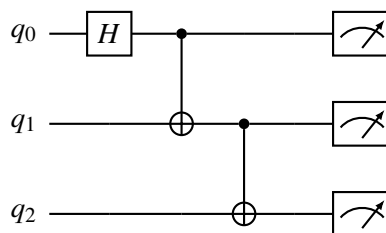
$$M = \begin{pmatrix} P(0|0) & P(0|1) \\ P(1|0) & P(1|1) \end{pmatrix}$$

Exercise 1. measurement error mitigation.

1. create a noise model that only has measurement errors:
 - $P(\text{measure } 0 | \text{state is } 1) = 0.05$ (5% chance of 1 being read as 0)
 - $P(\text{measure } 1 | \text{state is } 0) = 0.03$ (3% chance of 0 being read as 1)
2. build the calibration matrix by:
 - preparing $|0\rangle$ and measuring many times $\rightarrow$ gives column 0 of M
 - preparing $|1\rangle$ and measuring many times $\rightarrow$ gives column 1 of M
3. now prepare the state $|+\rangle$ and measure. compare:
 - ideal result (no noise)
 - noisy result (with measurement error)
 - mitigated result (apply M^{-1})
4. extend this to 2 qubits. how big is the calibration matrix now? how many calibration circuits do u need?

Exercise 2. combining noise and mitigation. this one puts everything together:

1. build a 3-qubit GHZ circuit
2. add a realistic noise model: depolarizing ($p = 0.005$ single, $p = 0.02$ CNOT) plus measurement error (3%)
3. run and record the raw noisy results
4. apply measurement error mitigation
5. apply ZNE (noise factors 1, 3, 5)
6. compare all four results: ideal, noisy, measurement-mitigated only, ZNE + measurement-mitigated
7. which technique helps more? does combining them help even more?



7 noise and circuit depth

Exercise 1. depth vs fidelity. this exercise builds intuition for why shallow circuits r so important in quantum ML:

1. build circuits of increasing depth: create a random circuit using repeated layers of $R_y(\theta)$ on each qubit followed by a CNOT entangling layer. use 4 qubits.

2. run each circuit with a realistic noise model and compute the output state fidelity.
3. plot fidelity vs circuit depth (number of layers).
4. at wt depth does fidelity drop below 0.5? below 0.9?
5. now imagine u r designing a variational circuit for ML. based on ur findings, wt's a reasonable maximum depth?

this is one of the key constraints in quantum ML: ur variational circuits can't be too deep or the noise will destroy all the information. keep this in mind for lab 5.

8 submission

submit a jupyter notebook with:

- all exercises completed
- plots comparing ideal vs noisy vs mitigated results
- the depth vs fidelity analysis
- a short reflection: how does noise affect the feasibility of quantum ML on near-term devices?